

DS Unit 5 — Sorting, Searching & Hashing

Complete Notes

Data Structures (DS) · Unit 5 · Sorting, Searching & Hashing · 2025

Overview

These notes cover **Sorting, Searching & Hashing**. Core data structures and the algorithms built on them — arrays, lists, stacks, trees, graphs, sorting and hashing. Each section states the core idea, a worked intuition and the points examiners look for.

1. Binary Search

Binary search halves a sorted range each step, running in $O(\log n)$.

Why it matters: Binary Search appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of binary search in one line.
- A small example showing binary search in practice.
- Common mistakes students make and how to avoid them.

2. Quick Sort

Quick sort partitions around a pivot; average $O(n \log n)$, worst case $O(n^2)$.

Why it matters: Quick Sort appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of quick sort in one line.
- A small example showing quick sort in practice.
- Common mistakes students make and how to avoid them.

3. Merge Sort

Merge sort divides the array, sorts halves recursively and merges them in $O(n \log n)$.

Why it matters: Merge Sort appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of merge sort in one line.
- A small example showing merge sort in practice.
- Common mistakes students make and how to avoid them.

4. Heap Sort

Heap sort builds a max-heap and repeatedly extracts the maximum in $O(n \log n)$ time.

Why it matters: Heap Sort appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of heap sort in one line.
- A small example showing heap sort in practice.
- Common mistakes students make and how to avoid them.

5. Hash Functions

Division, multiplication and universal hashing map keys to table indices.

Why it matters: Hash Functions appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of hash functions in one line.
- A small example showing hash functions in practice.
- Common mistakes students make and how to avoid them.

6. Collision Resolution

Chaining, linear probing, quadratic probing and double hashing resolve hash collisions.

Why it matters: Collision Resolution appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of collision resolution in one line.
- A small example showing collision resolution in practice.
- Common mistakes students make and how to avoid them.

Quick revision

[Binary Search](#) · [Quick Sort](#) · [Merge Sort](#) · [Heap Sort](#) · [Hash Functions](#) · [Collision Resolution](#)